

Answer_Assignment_VOLUME & SURFACE AREA

1. Ans. 21.49m, 900m², 1,800m³ ;

$$\text{Length of a diagonal} = \sqrt{l^2 + b^2 + h^2} = \sqrt{15^2 + 12^2 + 10^2} =$$

$$= \sqrt{225 + 144 + 100} = \sqrt{469} = 21.49\text{m},$$

$$\text{Surface area} = 2(lb + bh + lh) =$$

$$2(15\text{m} \times 12\text{m} + 12\text{m} \times 10\text{m} + 15\text{m} \times 10\text{m}) = 2(180 + 120 + 150) =$$

$$900\text{m}^2$$

$$\text{volume of the cuboid} = l \times b \times h = 15\text{m} \times 12\text{m} \times 10\text{m} = 1,800\text{m}^3$$

2. Ans. 20.97 m, 856m², 1,680m³;

$$\text{Length of a diagonal} = \sqrt{l^2 + b^2 + h^2} = \sqrt{10^2 + 12^2 + 14^2} =$$

$$= \sqrt{100 + 144 + 196} = \sqrt{440} = 20.97\text{m}$$

$$\therefore \text{Surface area} = 2(lb + bh + lh) =$$

$$2(10\text{m} \times 12\text{m} + 12\text{m} \times 14\text{m} + 10\text{m} \times 14\text{m}) =$$

$$2(120 + 168 + 140) = 856\text{m}^2$$

$$\therefore \text{Volume of a cuboid} = l \times b \times h = 10\text{m} \times 12\text{m} \times 14\text{m} = 1,680\text{m}^3$$

3. Ans. 14√3, 1,176m², 2,744 m³;

$$\text{Length of diagonal} = \sqrt{3} a = \sqrt{3} \times 14 = 14\sqrt{3},$$

$$\text{Surface area} = 6a^2 = 6 \times 14 \times 14 = 1176 \text{ and}$$

$$\text{Volume of a cube} = a^3 = 14^3 = 2744 \text{ m}^3$$

4. Ans. 16√3, 1,536m², 4,096 m³;

$$\text{Length of diagonal} = \sqrt{3} a = \sqrt{3} \times 16 = 16\sqrt{3},$$

$$\text{Surface area} = 6a^2 = 6 \times 16 \times 16 = 1,536 \text{ and}$$

$$\text{Volume of the cube} = a^3 = 16^3 = 4,096 \text{ m}^3$$

5. Ans. 1,728 m³;

$$\text{Surface area} = 6a^2 \Rightarrow 864 = 6a^2 \Rightarrow 144 = a^2 \therefore a = 12 \text{ m}$$

$$\therefore \text{Volume of the cube} = a^3 = 12^3 = 1728 \text{ m}^3$$

6. Ans. 2,197 m³;

$$\text{Surface area} = 6a^2 \Rightarrow 1014 = 6a^2 \Rightarrow 169 = a^2$$

$$\therefore a = 13 \text{ m}$$

$$\therefore \text{Volume of the cube} = a^3 = 13^3 = 2,197 \text{ m}^3$$

7. Ans. 512 m³;

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$$\text{Length of diagonal} = \sqrt{3} a \quad \Rightarrow 8\sqrt{3} = \sqrt{3} a$$

$$\therefore a = 8 \text{ m}$$

$$\therefore \text{Volume of the cube} = 8^3 = 512 \text{ m}^3$$

8.Ans.729 m³;

$$\text{Length of diagonal} = \sqrt{3} a \quad \Rightarrow 9\sqrt{3} = \sqrt{3} a$$

$$\therefore a = 9 \text{ m}$$

$$\therefore \text{Volume of a cube} = 9^3 = 729 \text{ m}^3$$

9.Ans. 336 m³;

$$\text{volume of a cuboid} = l \times b \times h = 8 \text{ m} \times 7 \text{ m} \times 6 \text{ m} = 336 \text{ m}^3$$

10.Ans.360 m³;

$$\text{volume of a cuboid} = l \times b \times h = 9 \text{ m} \times 8 \text{ m} \times 5 \text{ m} = 360 \text{ m}^3$$

11.Ans. 512 cm³;

$$\text{Volume of the cube} = a^3 = 8^3 = 512 \text{ m}^3$$

12.Ans. 729 cm³;

$$\text{Volume of a cube} = a^3 = 9^3 = 729 \text{ m}^3$$

13.Ans.4312 cm³, 1232 cm², 1540 cm²;

$$\text{Volume} = (\pi r^2 h) = \frac{22}{7} \times 7 \times 7 \times 28 = 4312 \text{ cm}^3,$$

$$\text{Curved surface area} = (2\pi rh) \Rightarrow 2 \times \frac{22}{7} \times 7 \times 28 = 1232 \text{ cm}^2 \text{ and Total surface area} = (2\pi rh + 2\pi r^2) =$$

$$(2 \times \frac{22}{7} \times 7 \times 28 + 2 \times \frac{22}{7} \times 7 \times 7) =$$

$$1232 + 308 = 1,540 \text{ cm}^2$$

14.Ans.960m³, 728m²;

$$\text{Volume of a cuboid} = l \times b \times h = 15 \text{ m} \times 16 \text{ m} \times 4 \text{ m} = 960 \text{ m}^3,$$

$$\text{Surface area} = 2(lb + bh + lh) =$$

$$2(15 \text{ m} \times 16 \text{ m} + 16 \text{ m} \times 4 \text{ m} + 15 \text{ m} \times 4 \text{ m}) = 2(240 + 64 + 60) =$$

$$728 \text{ m}^2.$$

15.Ans.960m³, 592m²;

$$\text{Volume of a cuboid} = l \times b \times h = 12 \text{ m} \times 10 \text{ m} \times 8 \text{ m} = 960 \text{ m}^3,$$

$$\text{Surface area} = 2(lb + bh + lh) =$$

$$2(12 \text{ m} \times 10 \text{ m} + 10 \text{ m} \times 8 \text{ m} + 12 \text{ m} \times 8 \text{ m}) = 2(120 + 80 + 96) =$$

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592m².

16.Ans.41,217 cm³;

Internal volume = $110 \times 70 \times 30 \text{ cm}^3 = 2,31,000\text{cm}^3$

External volume = $(110 + 2 \times 1.5) \times (70 + 2 \times 1.5) \times (30 + 2 \times 1.5) = 113 \times 73 \times 33 = 2,72,217 \text{ cm}^3$

∴ Volume of wood = external volume – internal volume =

41,217cm³

17.Ans.63,750 cm³;

Internal volume = $120 \times 60 \times 25 \text{ cm}^3 = 1,80,000\text{cm}^3$

External volume = $(120 + 2 \times 2.5) \times (60 + 2 \times 2.5) \times (25 + 2 \times 2.5) =$

$125 \times 65 \times 30 = 2,43,750 \text{ cm}^3$

∴ Volume of wood = external volume – internal volume = 63,750cm³

18.Ans.4 cms;

Volume of water = Volume of the field with rainwater as height

$$\Rightarrow 6000 = 500 \times 300 \times h \Rightarrow h = \frac{6000}{500 \times 300} = \frac{1}{25} \text{ m} = 4 \text{ cm}$$

19.Ans.10 cms;

Volume of water = Volume of the field with rainwater as height $\Rightarrow 8000 = 400 \times 200 \times h \Rightarrow h = \frac{8000}{400 \times 200} = \frac{1}{10} \text{ m} = 10 \text{ cm}$

20.Ans.6 cms; Volume of the new cube = $(3^3 + 4^3 + 5^3) = 216\text{cm}^3$

∴ Edge of the new cube = $\sqrt[3]{216} = 6 \text{ cm}$.

21.Ans: 9 cms;

Volume of the new cube = $(1^3 + 6^3 + 8^3) = 729\text{cm}^3$

Edge of the new cube = $\sqrt[3]{729} = 9 \text{ cm}$.

22. Ans.9:16;

Let their edges be a & b then, $\frac{a^3}{b^3} = \frac{27}{64} \Rightarrow \frac{a}{b} = \frac{3}{4}$

$\Rightarrow$ Ratio of their surface areas = $\frac{6a^2}{6b^2} = \frac{9}{16}$ i.e 9 : 16

23.Ans.25:36;

Let their edges be a & b then, $\frac{a^3}{b^3} = \frac{125}{216} \Rightarrow \frac{a}{b} = \frac{5}{6}$

$\Rightarrow$ Ratio of their surface areas = $\frac{6a^2}{6b^2} \Rightarrow \frac{25}{36}$ i.e 25 : 36

24.Ans.21 m;

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Let the depth of the tank be h metres.

$$\text{Then, } \pi \times (8)^2 \times h = 4224$$

$$\Rightarrow h = 4224 \times \frac{7}{22} \times \frac{1}{8 \times 8} = 21 \text{ m.}$$

25. Ans.63 m;

Let the depth of the tank be h metres.

$$\text{Then, } \pi \times (6)^2 \times h = 7128$$

$$\Rightarrow h = 7128 \times \frac{7}{22} \times \frac{1}{6 \times 6} = 63 \text{ m.}$$

26. Ans.3:14;

Let the radii of the cylinders be $2x$, $7x$ and their heights be $3y$, $4y$ respectively.

$$\text{Then, Ratio of their curved surface areas} = \frac{2\pi \times 2x \times 3y}{2\pi \times 7x \times 4y} = 3:14$$

27. Ans.6:7;

Let the radii of the cylinders be $10x$, $7x$ and their heights be $3y$, $5y$ respectively.

$$\text{Then, ratio of their curved surface areas} = \frac{2\pi \times 10x \times 3y}{2\pi \times 7x \times 5y} = 6:7$$

28. Ans.25m, 1,232m³, 550 cm², 704 cm²;

Here, $r = 7$ cm and $h = 24$ cm

$$\therefore \text{Slant height, } l = \sqrt{r^2 + h^2} = \sqrt{7^2 + 24^2} = \sqrt{625}$$

= 25 cm,

$$\text{Volume} = \frac{1}{3} \pi r^2 h = \left(\frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 24 \right) \text{ cm}^3 = 1232 \text{ cm}^3,$$

$$\text{Curved surface area} = \pi r l = \left(\frac{22}{7} \times 7 \times 24 \right) \text{ cm}^2 = 550 \text{ cm}^2 \text{ and}$$

$$\text{Total surface area} = (\pi r l + \pi r^2) = \left(550 + \frac{22}{7} \times 7 \times 7 \right) \text{ cm}^2 = 704 \text{ cm}^2$$

29. Ans.1,000m;

Here, $r = 7$ cm and $h = 24$ cm

$$\therefore \text{Slant height, } l = \sqrt{r^2 + h^2} = \sqrt{7^2 + 24^2} = \sqrt{625} = 25 \text{ m}$$

Area of the tent = Area of canvas

$$= \pi r l = \left(\frac{22}{7} \times 7 \times 25 \right) \text{ m}^2 = 2,310 \text{ m}^2$$

$$\therefore \text{Length of canvas} = \frac{\text{Area}}{\text{Width}} = \frac{2310}{2.31} = 1,000 \text{ m.}$$

30. Ans.13 m;

$$\text{Length of the longest pole} = \sqrt{l^2 + b^2 + h^2}$$

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$$= \sqrt{12^2 + 3^2 + 4^2} = \sqrt{144 + 9 + 16} = \sqrt{169} = 13 \text{ m}$$

31. Ans. 35 m;

$$\text{Length of the longest pole} = \sqrt{l^2 + b^2 + h^2}$$

$$= \sqrt{15^2 + 18^2 + 26^2} = \sqrt{225 + 324 + 676} = \sqrt{1225}$$

$$= 35 \text{ m}$$

32. Ans. Rs. 66,000 m;

Here, $r = 7 \text{ cm}$ and $h = 24 \text{ cm}$

$$\therefore \text{Slant height, } l = \sqrt{r^2 + h^2} = \sqrt{21^2 + 28^2} = \sqrt{1225} = 35 \text{ m}$$

$$\text{Area of the tent} = \text{Area of canvas} = \pi r l = \left(\frac{22}{7} \times 21 \times 35\right) \text{ m}^2 = 2310 \text{ m}^2$$

$$\therefore \text{Length of canvas} = \frac{\text{Area}}{\text{Width}} = \frac{2310}{1.75} = 1320 \text{ m.}$$

$$\therefore \text{Cost of canvas} = 1,320 \times 50 = \text{Rs. } 66,000$$

33. Ans. 45,000.

$$\text{No. of bricks} = \frac{90\% \text{ of volume of wall}}{\text{volume of brick}} = \frac{2400 \times 800 \times 60 \times 90}{24 \times 12 \times 8 \times 100} = 45,000$$

34. Ans. 486 cm²;

$$\text{Volume of the new cube} = (1^3 + 6^3 + 8^3) = 729 \text{ cm}^3$$

$$\therefore \text{Edge of the new cube} = \sqrt[3]{729} = 9 \text{ cm.}$$

$$\therefore \text{Surface area of the new cube} = 6a^2 = 6 \times 9 \times 9 = 486 \text{ cm}^2$$

35. Ans. 405;

$$\frac{\text{Vol. of cylinder}}{\text{Vol. of sphere}} = \frac{\pi \times 15 \times 15 \times 300}{\frac{4}{3} \times \pi \times 5 \times 5 \times 5} = \frac{15 \times 15 \times 300 \times 3}{4 \times 5 \times 5 \times 5} = 405$$

36. Ans. $\frac{2}{3} \text{ m}$;

$$\text{Volume of the soil dug out} = \pi r^2 h$$

$$\Rightarrow \left(\frac{22}{7} \times 10 \times 10 \times 10\right) = 3,142.85 \text{ m}^3.$$

Area of embankment = Area of embankment with well - Area of embankment without well

$$= \left(\frac{22}{7} \times 40 \times 40\right) - \left(\frac{22}{7} \times 10 \times 10\right) = 4,714 \text{ m}^2$$

$$\text{Height of embankment} = \frac{\text{Volume}}{\text{Area}} = \frac{3142.85}{4714.3} = \frac{2}{3} \text{ m.}$$

37. Ans. $\frac{1}{3} \text{ m}$;

$$\text{Volume of the soil dug out} = \pi r^2 h$$

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$$\Rightarrow \left(\frac{22}{7} \times 7 \times 7 \times 8\right) = 1,232\text{m}^3.$$

Area of embankment = Area of embankment with well - Area of embankment without well

$$\Rightarrow \left(\frac{22}{7} \times 35 \times 35\right) - \left(\frac{22}{7} \times 7 \times 7\right) = 3696 \text{ m}^2$$

$$\text{Height of embankment} = \frac{\text{Volume}}{\text{Area}} = \frac{1232}{3696} = \frac{1}{3} \text{ m}.$$

38. Ans.15cm approx;

Let the radius of the ball be r cm.

Volume of the ball = Volume of water displaced by it

$$\therefore \frac{4}{3} \pi r^3 = \frac{22}{7} \times 21 \times 21 \times 10.2 \Rightarrow r = 15 \text{ cm approx.}$$

39. Ans. 21cm;

Let the radius of the ball be r cm.

Volume of the ball = Volume of water displaced by it

$$\therefore \frac{4}{3} \pi r^3 = \frac{22}{7} \times 28 \times 28 \times 15.75$$

$$\Rightarrow r^3 = 9261 \Rightarrow r = 21 \text{ cm}.$$

40. Ans.6%(approx.);

Wood wasted = vol. of the rectangular block – vol. of the cone

$$= 10 \times 8 \times 7 - \frac{1}{3} \times \frac{22}{7} \times 6 \times 6 \times 14 = 560 - 528 = 32$$

$$\therefore \% \text{ of wood wasted} = \frac{32}{560} \times 100 = 6\% \text{ (approx.)}$$